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Graduation Project Proposal

<REPLICA>

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1 Introduction

Alzheimer's disease does not begin on the day a patient receives a diagnosis. Changes in the brain associated with Alzheimer's disease (AD) can begin a decade or more before noticeable problems with memory and thinking appear [1]. This long and gradual disease process makes Alzheimer's particularly difficult to monitor because a patient may be changing biologically and cognitively long before those changes become obvious in daily life. Alzheimer's disease is the most common cause of dementia, contributing to approximately 60–70% of dementia cases. In 2021, an estimated 57 million people were living with dementia worldwide, highlighting the growing impact of cognitive disorders on patients, families, healthcare systems, and communities [2].

Understanding how an individual patient is changing over time is therefore an important part of Alzheimer's research and care. Cognitive status is not the same for every patient. Some individuals remain cognitively unimpaired, while others develop Mild Cognitive Impairment (MCI), a condition in which memory or thinking difficulties are greater than expected for age but are not yet severe enough to prevent normal daily activities [3]. Importantly, MCI does not always progress to dementia; some individuals remain stable and some may even improve [3].

Among patients who do decline, the speed and pattern of decline can differ substantially. A longitudinal study of 936 individuals with MCI from the Alzheimer's Disease Neuroimaging Initiative (ADNI), for example, identified four distinct cognitive trajectories ranging from almost no decline to aggressive decline [4]. This means that two patients who appear similar during one clinical visit may follow very different paths in the following months or years.

This variability creates the central problem addressed by this project: a single clinical visit can describe a patient's current condition, but it cannot by itself show the direction or speed of that patient's cognitive change. Information relevant to Alzheimer's progression may be collected across several visits and may include cognitive assessments, demographic characteristics, neuroimaging, genetic information, and biological markers. Looking at these measurements longitudinally allows changes and patterns to be captured rather than treating every examination as an isolated snapshot.

Evidence supports the value of this approach. In a study using ADNI data, Ding et al. compared a model based on several visits with a model based only on a single examination and found that incorporating longitudinal, multimodal information improved the prediction of progression from MCI to Alzheimer's disease [5]. ADNI is particularly suitable for this type of research because it is a longitudinal, multicenter study containing repeated clinical, cognitive, imaging, genetic, and biomarker measurements from participants at different stages of cognitive impairment [6].

2 The Problem

For patients with Mild Cognitive Impairment (MCI), one of the major challenges is that their future cognitive course cannot be understood from their current cognitive status alone. Patients who appear similar at one point in time may subsequently follow substantially different trajectories.

In a longitudinal study of 936 individuals with MCI from the Alzheimer's Disease Neuroimaging Initiative (ADNI), four distinct cognitive trajectories were identified over five years: approximately 27% showed almost no cognitive change, whereas other groups experienced mild, moderate, or aggressive decline [4]. This heterogeneity makes it difficult to determine from a single assessment how quickly an individual patient's cognitive function is changing or where that trajectory may lead.

Consider two patients with MCI who obtain the same MMSE score during their current assessment. Although their scores are identical at that visit, their previous measurements may reveal very different patterns: one patient's score may have remained relatively stable across previous visits, while the other may have been declining progressively.

Treating the current scores as isolated observations would therefore hide an important difference between the two patients; the direction and rate of cognitive change over time. This distinction is clinically relevant because longitudinal studies of MCI have demonstrated substantial variation in cognitive trajectories. In the ADNI study of 936 individuals with MCI, four distinct trajectories were identified over five years, ranging from nearly no cognitive change to mild, moderate, and aggressive decline [4].

The challenge is not simply the availability of patient information, but how to use measurements collected across time to estimate an individualized future cognitive trajectory. Relevant information may include repeated cognitive assessments together with demographic, genetic, and imaging-derived measurements, and these observations may occur at different points throughout follow-up.

Evidence suggests that preserving this temporal information is valuable: Ding et al. found that a model incorporating longitudinal multimodal measurements from multiple visits achieved better prediction of MCI-to-Alzheimer's progression than a model based on a single examination [5].

Therefore, the specific problem addressed in this project is the limited ability of a single-timepoint view to represent and anticipate an individual MCI patient's cognitive change over time. The project focuses on transforming longitudinal patient history into a personalized estimate of future MMSE trajectory, so that differences in the direction and rate of cognitive change can be represented at the individual-patient level rather than relying only on the patient's current score or a population-level pattern.

3 The Solution

REPLICA attempts to solve the problem through a web-based platform that follows each patient's cognitive condition across multiple visits. The platform creates a personalized Digital Twin that combines the patient's previous MMSE scores with relevant personal and clinical information to predict a future MMSE trajectory [7].

The system will use longitudinal data from ADNI. Selected variables will include the actual month of each visit, MMSE scores, age, sex, education, APOE4 status, and selected numerical brain measurements such as hippocampal volume [6]. The project will use measurements already extracted from MRI scans rather than processing raw MRI images.

A Linear Mixed-Effects Model (LMM) will form the main prediction model. LMM is a statistical model designed for repeated measurements, allowing it to capture both fixed effects, which represent general patterns shared across the population, and random effects, which represent individual differences between patients. In REPLICA, it will learn the general pattern of MMSE change across the study population while also estimating a patient-specific starting level and rate of change from the visits available for that patient. Because time is entered as an actual numerical value, the model can work with irregularly spaced visits [8].

Personalization will become stronger as more visits are available; for a patient with only one visit, the initial estimate will rely more heavily on population patterns and patient characteristics. A Gradient Boosting model, such as XGBoost, will then learn to correct the remaining prediction error using variables such as age, education, APOE4 status, hippocampal volume, and the prediction produced by the mixed-effects model [9]. The final output will be an estimated MMSE score at selected future months and a personalized curve showing the expected direction and rate of cognitive change.

Because the project will use historical rather than live hospital data, the Digital Twin update process will be demonstrated through sequential simulation. A patient's historical visits will be processed chronologically as though each visit has newly become available. After each new visit, the Digital Twin will update its current state while preserving previous states. This will demonstrate how the Digital Twin could evolve when new patient data is received in a clinical environment.

REPLICA will therefore represent the patient as a dynamic entity rather than generating one isolated result. For example, after early visits with MMSE scores of 28 and 27, the Twin may predict a score of 26 at month 12 and 24 at month 24. If a later observed score is lower than expected, the Twin will update the personalized curve and may show a faster expected decline. The numbers in this example are illustrative; actual predictions will be produced by the trained models.

The clinician will be able to review these changes as a timeline rather than viewing only the newest prediction. The centralized dashboard will allow physicians to view their monitored patient list, open an individual patient's Digital Twin, examine historical MMSE and clinical measurements, compare observed and predicted scores, inspect the expected trajectory, and identify patients whose predicted decline may require attention.

The main benefit of REPLICA is that it converts separate visits into one understandable, patient-specific trajectory. It can help physicians recognize gradual decline, compare patients with similar current scores but different histories, and prioritize closer follow-up for patients with a faster predicted decline. The use of actual visit times also allows the model to handle irregular follow-up schedules, while the dashboard makes the results easier to review without examining separate records or raw model outputs.

REPLICA is intended to support clinician interpretation and follow-up. It will not autonomously diagnose Alzheimer's disease or make treatment decisions.

4 Product Vision

For clinicians and neurologists monitoring patients with Mild Cognitive Impairment (MCI). Who need an early, personalized indication of how a patient's condition is likely to progress, rather than only a snapshot of where they stand today. REPLICA is a patient-specific Alzheimer's digital twin platform. That maintains an evolving representation of each patient's condition, using their clinical history, assessments, and genetic risk factors to recognize patient-specific patterns of decline and simulate likely future trajectories. Unlike traditional monitoring approaches that primarily rely on single-visit snapshots and general population reference ranges to classify a patient's current stage [9]. Our product combines patient-specific prediction with an evolving simulation, adapting to each patient's own pattern of change through their visit history.

5 Product Roadmap

Sprint	Target date	Planned work
Sprint 0	27 September	Requirements analysis; ADNI data access request; literature review; initial system design
Sprint 1	22 October	Data cleaning and merging pipeline, feature engineering, baseline model development
Sprint 2	19 November	Baseline model training and evaluation, working dashboard
Release 1		
Sprint 3	TBA	Personalized Digital Twin core: Mixed-Effects Model development and training
Sprint 4	TBA	Gradient boosting correction layer (XGBoost), full model evaluation
Sprint 5	TBA	Full clinician dashboard integration; testing; final report
Release 2		

6 Objectives

The objectives of this project are defined across three levels (product, project, and learning) to capture not only what the system aims to achieve for its users, but also the development process required to build it and the technical growth gained by completing it.

6.1.1 Product Objectives (Costumer Focus – Plan)

1. To estimate an individual MCI patient's future MMSE trajectory and expected rate of cognitive decline, addressing the limitation of conventional single-timepoint cognitive assessment, which primarily reflects a patient's current cognitive state rather than anticipating how it may change over time. Clinicians benefit from earlier, more individualized awareness of a patient's likely cognitive progression.

2. To provide a personalized Digital Twin for each patient that maintains and updates an evolving representation of the patient's cognitive state as additional clinical visits become available.

3. To identify patients who may require greater clinical attention based on their predicted rate of cognitive decline and changes in their historical cognitive state.

4. To present model outputs through a clear clinical dashboard, solving the problem that raw AI/ML outputs are not directly usable by clinicians without data-science expertise. Clinicians benefit from being able to interpret a patient's trajectory and predicted cognitive decline without needing to understand the underlying model.

5. To preserve a historical timeline of each patient's Digital Twin states so clinicians can understand both the current state and how it has changed across previous visits.

6. To demonstrate continuous Digital Twin updating through simulation by processing historical patient visits sequentially, while excluding direct connection to live hospital systems or medical devices from the current project scope.

6.1.2 Project Objectives (Solution Focus – Plan)

1. To acquire and preprocess the ADNI/ADNIMERGE2 dataset, linking diagnosis, cognitive, genetic, and MRI-derived tables.

2. To construct longitudinal per-patient sequences (MMSE, CDR-SB, visit timing) with a patient-level train/test split to avoid data leakage.

3. To apply a Linear Mixed Model (LMM) to model longitudinal changes in patients' cognitive and clinical measurements while accounting for both population-level trends and individual patient differences.

4. To extract meaningful longitudinal features from the LMM, such as estimated rates of cognitive change, individual patient trajectories, and other patient-specific progression characteristics.

5. To train and validate a Gradient Boosting model using baseline variables, LMM-derived longitudinal features, and LMM predictions to correct residual prediction errors and improve future MMSE trajectory forecasting.

6. To build a Digital Twin simulation layer that updates each patient's represented cognitive state as historical visits are processed sequentially and generates predicted future progression based on the patient's observed longitudinal characteristics.

7. To evaluate the models using regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 with patient-level validation and cross-validation, while comparing performance across the baseline and proposed LMM + Gradient Boosting pipeline.

8. To develop a clinical dashboard presenting predicted MMSE trajectories and historical patient states.

6.1.3 Learning Objectives (Student Focus)

1.To gain hands-on experience applying Linear Mixed Models to biomedical longitudinal data and understanding how statistical models can represent changes in patient measurements over time.

2.To learn how to work with large, multi-table clinical datasets, linking records by patient and visit identifiers while maintaining data consistency.

3. To learn multimodal feature integration techniques for combining clinical, cognitive, genetic, and imaging-derived features for Alzheimer's progression prediction.

4. To learn how to extract and interpret individual longitudinal trajectories from Linear Mixed Models, including population-level effects, patient-specific effects, and rates of change.

5. To gain practical experience handling small sample sizes, class imbalance, missing data, and irregular patient follow-up in medical ML.

6. To gain practical experience training and tuning Gradient Boosting models for clinical prediction and understanding their advantages for nonlinear relationships in tabular medical data.

7. To learn rigorous model validation techniques, including patient-level splitting, cross-validation, prevention of data leakage, and comparison against appropriate baseline models.

8. To learn how longitudinal statistical modeling and machine learning can be combined to transform repeated patient observations into meaningful personalized cognitive progression predictions.

9. To learn how to design an interactive dashboard that presents AI/ML predictions, longitudinal patient trajectories, and Digital Twin states to non-technical clinical users.

7 Scope

This project, developed within the resource, time, and data constraints of a bachelor's graduation project. Its scope is defined at a high level as follows:

-Technology/Platform Support: The system will be delivered as a web-based application accessible through standard modern browsers on desktop devices. Native mobile applications and embedded-device deployment are outside the scope of this project.

-Data Source: The system will use retrospective, de-identified data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), with OASIS considered as an accessible backup dataset. Live data streamed from hospital systems or wearable devices is outside the scope.

-Clinical Deployment: The system is a research prototype intended to demonstrate technical feasibility and is not a certified medical device. It is not intended for real-world clinical deployment, autonomous diagnosis, or direct treatment decisions.

-Model Update Mechanism: Patient-specific digital twin adaptation will be demonstrated through an offline simulation of sequential, historical visit batches. Live, continuous updating through an active clinical monitoring system is outside the scope.

-Imaging Scope: The MRI modality is represented through structural features already derived by standard FreeSurfer processing (hippocampal volume, with additional regional volumes such as the entorhinal cortex, amygdala, and ventricles included where available), not raw 3D MRI scans. Training a convolutional model directly on raw imaging volumes would require a separate raw-image data acquisition and is not part of this project's scope.

-Predictive Modeling Scope: Patient-level personalization is implemented through a hierarchical statistical model (a linear mixed-effects model capturing an individual random intercept and slope per patient around a population-level trend), combined with a gradient-boosted correction layer trained on each patient's static, genetic, and MRI-derived features. This design was chosen specifically for its ability to natively handle irregularly spaced clinical visits without requiring large-scale or GPU-dependent training. Model performance is evaluated against a naive per-patient linear extrapolation baseline, and any case where personalization does not outperform this baseline will be reported transparently along with its likely cause, rather than omitted or reframed.

-Language Support: The system interface and outputs will be provided in English, designed for clinical/technical users. Multi-language localization is outside the scope of this phase.

-AI Model Scope: The predictive system focuses exclusively on personalized cognitive progression forecasting through future MMSE trajectory prediction using longitudinal cognitive, genetic, and MRI-derived structural features. Diagnosis of Alzheimer’s disease, other dementia subtypes, and treatment recommendation are outside the scope.

8 Hardware/Software Tools and Cost

Hardware Tools	
Personal Laptops (team members)	0SAR (already owned)
Google Colab	0SAR
Google Drive (15 GB free)	0SAR
Software Tools	
Jupyter Notebook	0 SAR
Statsmodels	0 SAR
Scikit-learn	0 SAR
XGBoost	0 SAR
Git + GitHub	0 SAR
Visual Studio Code	0 SAR
Jira	0 SAR
Python	0 SAR
matplotlib, seaborn	0 SAR
Streamlit (interactive dashboard)	0 SAR
ADNI data access	0 SAR (requires approved research data-use application)

9 Scrum Team

9.1 Skill Set Requirements

Technical Skill Required	What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan
Python programming	Intermediate
Longitudinal and time-series modeling	Beginner
Gradient Boosting / XGBoost	Beginner Learning plan: XGBoost and scikit-learn official documentation and tutorials
Data preprocessing	Intermediate
Machine learning evaluation	Intermediate
REST API development (Flask / FastAPI)	Beginner Learning plan: Flask/FastAPI official documentation, YouTube tutorials
Frontend development	Intermediate
Database management	Beginner
Version control (Git / GitHub)	Intermediate
UI/UX design	Intermediate
Agile / Scrum methodology	Intermediate
Alzheimer's clinical-data interpretation	Beginner

	Conduct a literature review and consult domain experts

9.2 Roles and Responsibilities

Scrum Team	
Product Owner: (list name of the project supervisor)	Dr. Sharefah Alghamdi
Developers: (list student names)	Reem Almaskati Sadeem Al Sultan Sondos Aldughaim Danah AlDhowaihy
Scrum Master (SM): (list name of designated SM for the project)	Sondos Aldughaim
Stakeholders: (list names of project stakeholders, include customer names if any)	Project Supervisor- Physicians - Patients with MCI- Saudi Ministry of Health- King Saud Univesity- Team members

10 References

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